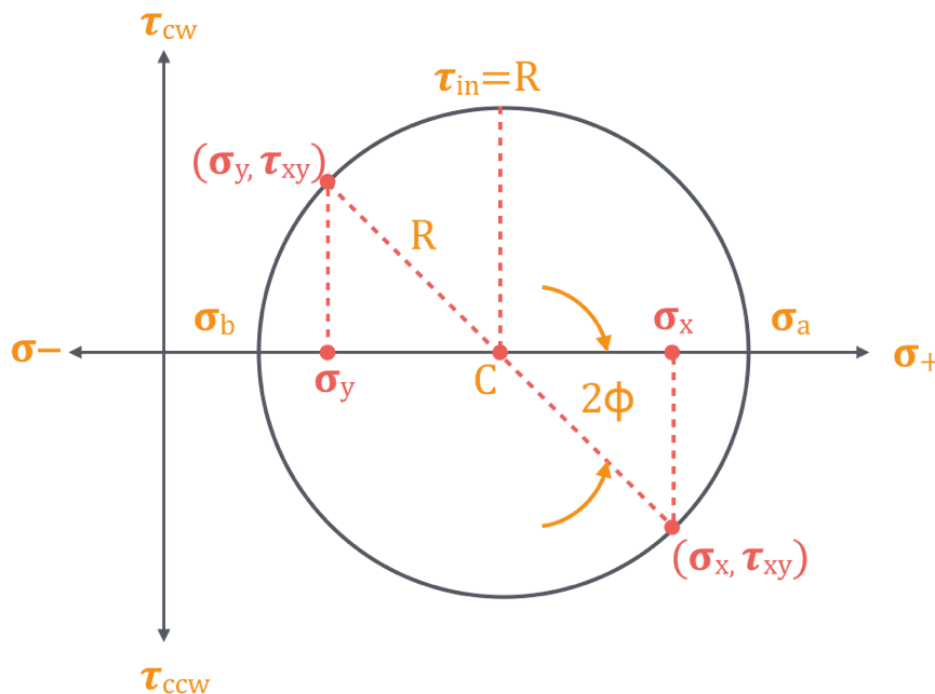


3 MIN READ

Engineers most often want to determine the maximum normal stress induced at a given point for a particular application or design.

Mohr's Circle is a graphical tool that is commonly used by engineers to graphically analyze the principal and maximum shear stresses on any plane, as well as provide graphical coordinates of these shear stresses.



However, there can be an infinite number of planes passing through a point, and the normal stress on each plane will vary.

The **PRINCIPAL PLANE** or maximum principal plane is the plane on which the normal stress value is at a MAXIMUM, with this value being referred to as the **MAXIMUM PRINCIPAL STRESS**.

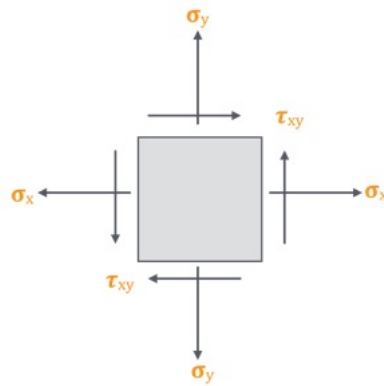
Similarly, there will be one more plane, known as the **PRINCIPAL PLANE**, on which the normal stress value is at a MINIMUM, with this value being referred to as the **MINIMUM PRINCIPAL STRESS**.

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The TOPIC of **MOHR'S CIRCLE** can be referenced under the SUBJECT of **MECHANICS OF MATERIALS** on page 81 of the **NCEES Supplied Reference Handbook**, Version 9.4 for Computer Based Testing.

Using a graphical approach, we are able to determine the PRINCIPAL STRESSES on each plane using a MOHR'S CIRCLE.

Mohr's circle is a geometric representation of the 2-D transformation of stresses, in which the component stresses and are found as the coordinates of a point whose location depends upon the angle to determine the aspect of the cross section.

The Mohr's circle is used to determine the principle angles (orientations) of the principal stresses without have to plug an angle into stress transformation equations.

To draw a Mohr's Circle for a typical 2-D element, we can use the following procedure to determine the principal stresses.

Define The Shear Stress Coordinate System:

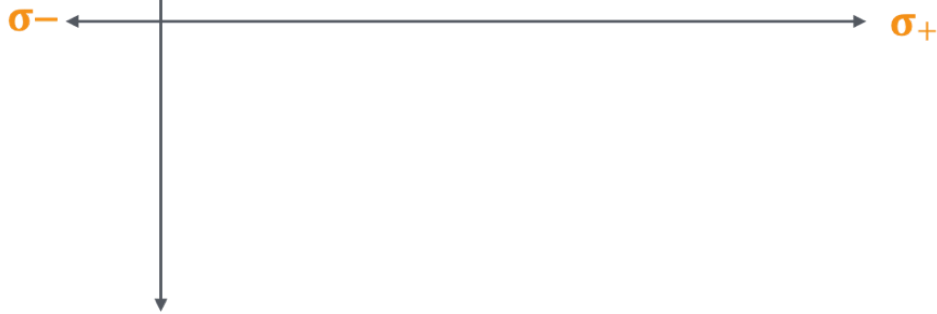
1. Define the coordinate system for the normal
Tensile normal stress components are plotted on the horizontal axis and are considered positive. Compressive normal stress components are also plotted on the horizontal axis and are negative.

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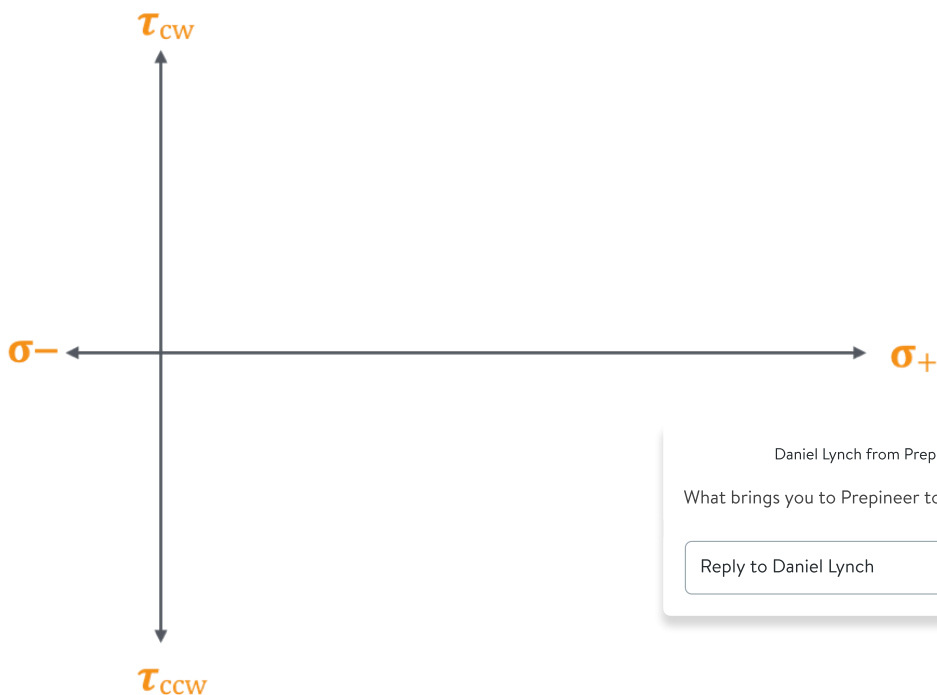
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Define The Torsional Coordinate System:

2. For the construction of a Mohr's circle, shearing stresses are plotted ABOVE the normal stress axis when the pair of shearing stresses, acting on opposite and parallel faces of an element, form a CLOCKWISE (cw) couple. Shearing stresses are plotted BELOW the normal axis when the shear stresses form a COUNTERCLOCKWISE (ccw) couple.



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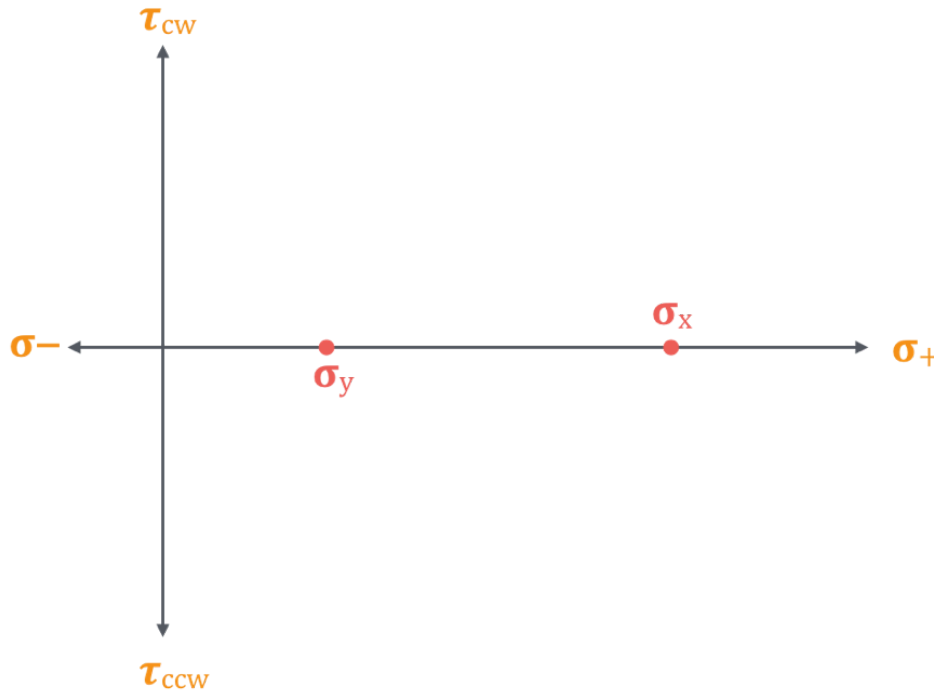
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Plot The Shear Stress Values:

3. Plot the shear stress values given in the problem statement, or plot generic points on the for σ_x -axis and σ_y as shown below.



Plot The Magnitude Of The Couple:

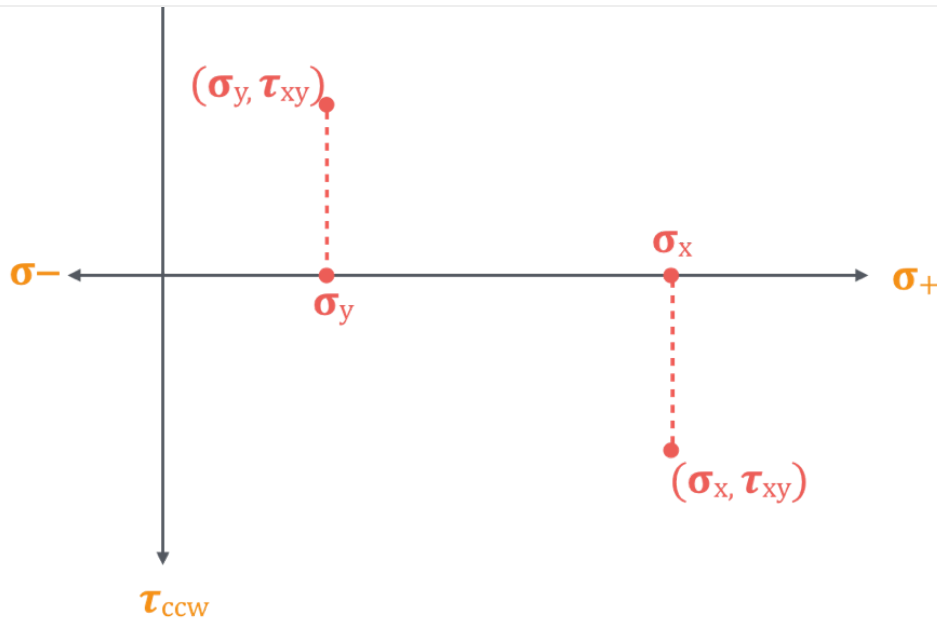
4. Plot the magnitude of the couple given in the problem statement with a clockwise (cw) couple being plotted above the σ_x -axis, and a counterclockwise (ccw) couple being plotted below the σ_x -axis. If not values are provided for the moment plot generic points above and below the σ -axis for τ_{xy} as shown below.

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Obtain The Center Of The Mohr's Circle:

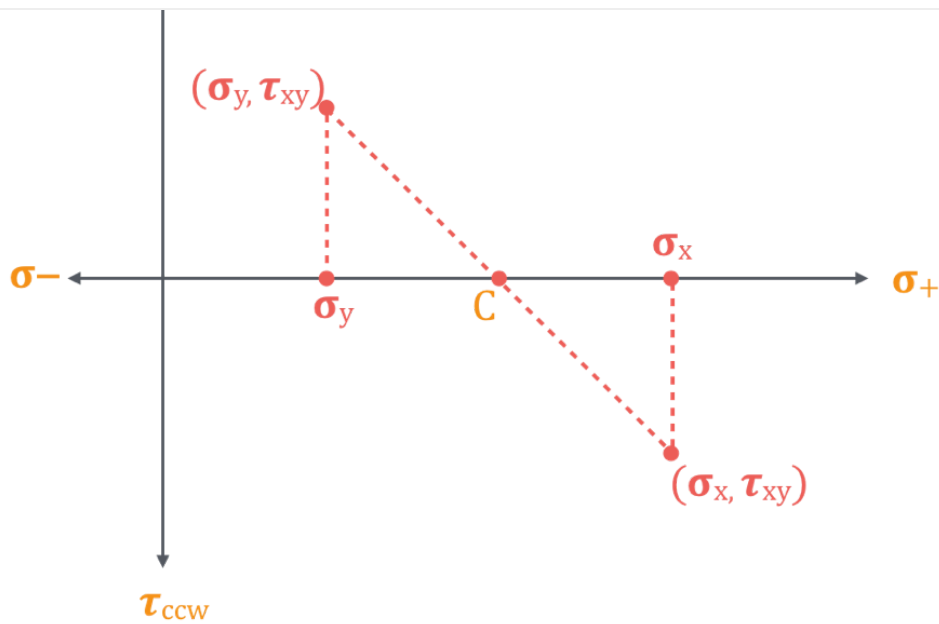
5. The center of the Mohr's circle is obtained graphically by plotting the two points representing the two known states of stress, and drawing a straight line between the two points. The intersection of this straight line and the σ -axis is the location of the center of the circle.

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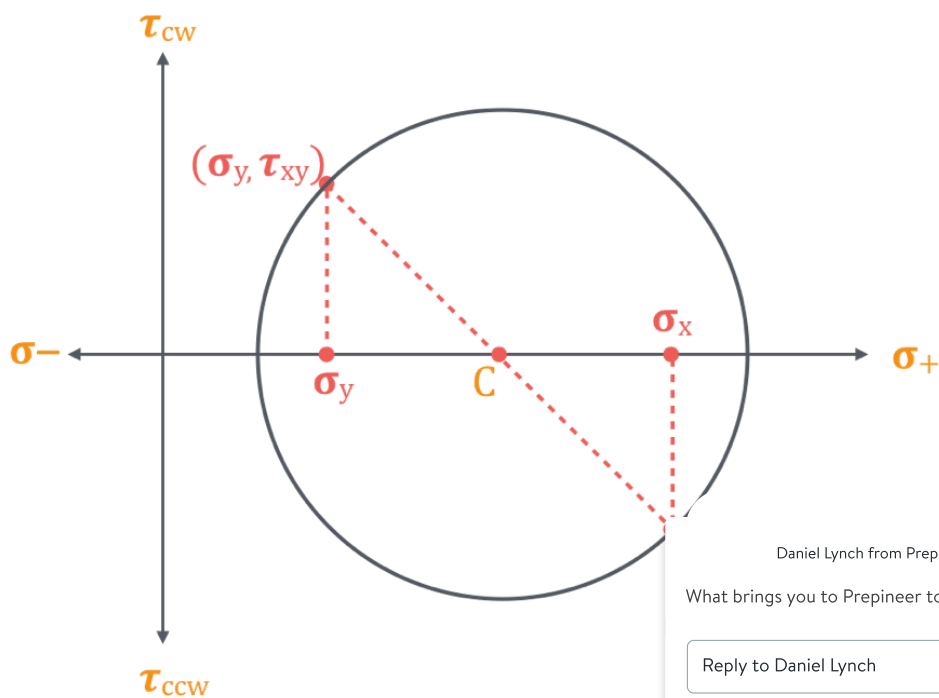
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Draw The Mohr's Circle:

6. Draw the Mohr's circle assuming the connection line as the diameter of the circle, using the intersection of the diagonal straight line and the σ -axis as the center of the circle.



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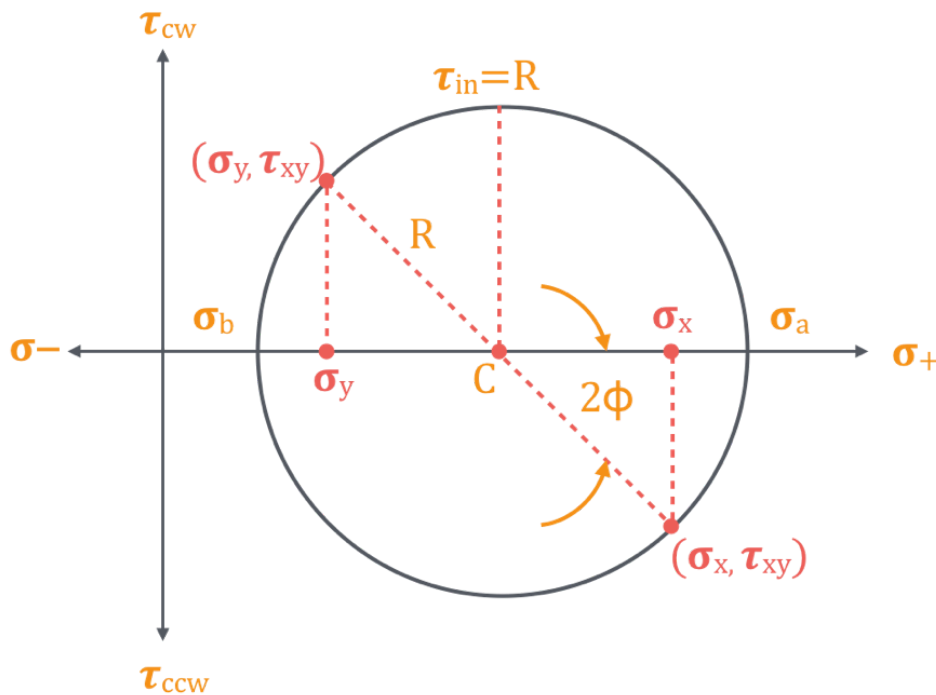
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Circle:

7. Stress Analysis on Mohr's circle – To get normal and shear stress values at any plane theta, take angle 2ϕ in the Mohr's circle starting from diagonal of the circle and locate a peripheral point as shown. Shear stress value will be on the y-axis and normal stress values will be on the x-axis.



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CHRIS LOPEZ



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